2.11 Series: Method of Differences - Part 2

Worked Solutions

1)

$$\frac{1}{r(r+1)} = \frac{A}{r} - \frac{B}{r+1},$$
$$1 = A(r+1) - Br.$$

When $r = -1$, $1 = B$. When $r = 0$, $1 = A$. Therefore $A = 1$, $B = 1$.

$$\begin{aligned}\sum_{r=1}^{n} \frac{1}{r(r+1)} &= \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1}\right) \\ &= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{n-1} - \frac{1}{n} + \frac{1}{n} - \frac{1}{n+1} \\ &= 1 - \frac{1}{n+1} = \frac{n}{n+1}.\end{aligned}$$

2)

$$\frac{1}{r(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+2},$$
$$1 = A(r+1)(r+2) + Br(r+2) + Cr(r+1).$$

When $r = 0$, $1 = 2A$; when $r = -1$, $1 = -B$; when $r = -2$, $1 = 2C$. Hence $A = \frac{1}{2}$, $B = -1$, $C = \frac{1}{2}$.

$$\begin{aligned}\sum_{r=1}^{40} \frac{1}{r(r+1)(r+2)} &= \sum_{r=1}^{40} \left(\frac{1}{2r} - \frac{1}{r+1} + \frac{1}{2(r+2)}\right) \\ &= \frac{1}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{82} - \frac{1}{41} + \frac{1}{84} = \frac{215}{861}.\end{aligned}$$

3)

$$\frac{1}{r^2+5r+6} = \frac{1}{(r+2)(r+3)} = \frac{A}{r+2} + \frac{B}{r+3},$$
$$1 = A(r+3) + B(r+2).$$

When $r = -3$, $1 = -B$; when $r = -2$, $1 = A$. Thus $A = 1$, $B = -1$, and $\frac{1}{(r+2)(r+3)} = \frac{1}{r+2} - \frac{1}{r+3}$.

$$\begin{aligned}\sum_{r=10}^{200} \frac{1}{(r+2)(r+3)} &= \sum_{r=10}^{200} \left(\frac{1}{r+2} - \frac{1}{r+3}\right) \\ &= \frac{1}{12} - \frac{1}{13} + \frac{1}{13} - \frac{1}{14} + \cdots + \frac{1}{202} - \frac{1}{203} \\ &= \frac{1}{12} - \frac{1}{203} = \frac{191}{2436}.\end{aligned}$$

4)

$$\frac{1}{(r+1)(r+2)(r+3)} = \frac{A}{r+1} + \frac{B}{r+2} + \frac{C}{r+3},$$
$$1 = A(r+2)(r+3) + B(r+1)(r+3) + C(r+1)(r+2).$$

When $r = -1$, $1 = 2A$; when $r = -2$, $1 = -B$; when $r = -3$, $1 = 2C$. Hence $A = \frac{1}{2}$, $B = -1$, $C = \frac{1}{2}$.

$$\sum_{r=5}^{55} \frac{1}{(r+1)(r+2)(r+3)} = \sum_{r=5}^{55} \left(\frac{1}{2(r+1)} - \frac{1}{r+2} + \frac{1}{2(r+3)} \right)$$
$$= \frac{1}{12} - \frac{1}{7} + \frac{1}{14} + \frac{1}{114} - \frac{1}{57} + \frac{1}{116} = \frac{136}{11571}.$$

5)

$$\frac{4}{(r+3)(r+4)} = \frac{A}{r+3} + \frac{B}{r+4},$$
$$4 = A(r+4) + B(r+3).$$

When $r = -4$, $4 = -B$; when $r = -3$, $4 = 4A$. Hence $A = 1$, $B = -1$, so $\frac{4}{(r+3)(r+4)} = \frac{1}{r+3} - \frac{1}{r+4}$.

$$\sum_{r=10}^{n} \frac{4}{(r+3)(r+4)} = \sum_{r=10}^{n} \left(\frac{1}{r+3} - \frac{1}{r+4} \right)$$
$$= \frac{1}{13} - \frac{1}{14} + \frac{1}{14} - \frac{1}{15} + \cdots + \frac{1}{n+3} - \frac{1}{n+4}$$
$$= \frac{1}{13} - \frac{1}{n+4} = \frac{n-9}{13(n+4)}.$$

6)

$$4r^2 - 4r - 3 = (2r+1)(2r-3),$$
$$\frac{4}{4r^2 - 4r - 3} = \frac{4}{(2r+1)(2r-3)} = \frac{A}{2r+1} + \frac{B}{2r-3},$$
$$4 = A(2r-3) + B(2r+1).$$

When $r = \frac{3}{2}$, $4 = 4B$; when $r = -\frac{1}{2}$, $4 = -4A$. Hence $A = -1$, $B = 1$, so $\frac{4}{4r^2-4r-3} = \frac{1}{2r-3} - \frac{1}{2r+1}$.

$$\sum_{r=2}^{n} \frac{4}{4r^2 - 4r - 3} = \sum_{r=2}^{n} \left(\frac{1}{2r-3} - \frac{1}{2r+1} \right)$$
$$= 1 - \frac{1}{5} + \frac{1}{3} - \frac{1}{7} + \frac{1}{5} - \frac{1}{9} + \cdots + \frac{1}{2n-3} - \frac{1}{2n+1}$$
$$= \frac{4}{3} - \frac{1}{2n-1} - \frac{1}{2n+1}$$
$$= \frac{4}{3} - \frac{2n+1}{(2n-1)(2n+1)} - \frac{2n-1}{(2n-1)(2n+1)}$$
$$= \frac{4(4n^2-1)}{3(2n-1)(2n+1)} - \frac{4n}{(2n-1)(2n+1)}$$
$$= \frac{4(4n^2-3n-1)}{3(4n^2-1)} = \frac{4(n-1)(4n+1)}{12n^2-3}.$$

7 a)

$$\begin{aligned} r^2+4r+3 &= (r+3)(r+1), \\ \frac{2}{(r+3)(r+1)} &= \frac{A}{r+3}+\frac{B}{r+1}, \\ 2 &= A(r+1)+B(r+3). \end{aligned}$$

When $r=-1$, $2=2B$; when $r=-3$, $2=-2A$. Thus $A=-1$, $B=1$ and $\frac{2}{r^2+4r+3}=\frac{1}{r+1}-\frac{1}{r+3}$.

$$\begin{aligned} \sum_{r=1}^{n}\frac{2}{r^2+4r+3} &= \sum_{r=1}^{n}\left(\frac{1}{r+1}-\frac{1}{r+3}\right) \\ &= \frac{1}{2}-\frac{1}{4}+\frac{1}{3}-\frac{1}{5}+\frac{1}{4}-\frac{1}{6}+\cdots+\frac{1}{n+1}-\frac{1}{n+3} \\ &= \frac{1}{2}+\frac{1}{3}-\frac{1}{n+2}-\frac{1}{n+3}=\frac{5}{6}-\frac{1}{n+2}-\frac{1}{n+3}. \end{aligned}$$

7 b)

$$\sum_{r=1}^{n}\frac{1}{r^2+4r+3}=\frac{1}{2}\left[\frac{5}{6}-\frac{1}{n+2}-\frac{1}{n+3}\right].$$

As $n\to\infty$, $\frac{1}{n+2}\to 0$ and $\frac{1}{n+3}\to 0$, so

$$\sum_{r=1}^{\infty}\frac{1}{r^2+4r+3}=\frac{5}{12}.$$

8)

$$\begin{aligned} \frac{1}{(2r+1)(2r+3)} &= \frac{A}{2r+1}+\frac{B}{2r+3}, \\ 1 &= A(2r+3)+B(2r+1). \end{aligned}$$

When $r=-\frac{3}{2}$, $1=-2B$; when $r=-\frac{1}{2}$, $1=2A$. Thus $A=\frac{1}{2}$, $B=-\frac{1}{2}$, and

$$\frac{1}{(2r+1)(2r+3)}=\frac{1}{2(2r+1)}-\frac{1}{2(2r+3)}.$$

$$\begin{aligned} \sum_{r=1}^{n}\frac{1}{(2r+1)(2r+3)} &= \sum_{r=1}^{n}\left(\frac{1}{2(2r+1)}-\frac{1}{2(2r+3)}\right) \\ &= \frac{1}{2\times 3}-\frac{1}{2\times 5}+\frac{1}{2\times 5}-\frac{1}{2\times 7}+\cdots+\frac{1}{2(2n+1)}-\frac{1}{2(2n+3)} \\ &= \frac{1}{6}-\frac{1}{2(2n+3)}. \end{aligned}$$

Therefore $\displaystyle\sum_{r=1}^{\infty}\frac{1}{(2r+1)(2r+3)}=\frac{1}{6}$.

9 a)

$$\frac{3}{r^2+3r} = \frac{3}{r(r+3)} = \frac{A}{r} + \frac{B}{r+3},$$
$$3 = A(r+3) + Br.$$

When $r = -3$, $3 = -3B$; when $r = 0$, $3 = 3A$. Thus $A = 1$, $B = -1$, so $\frac{3}{r^2+3r} = \frac{1}{r} - \frac{1}{r+3}$.

$$\begin{aligned}\sum_{r=1}^{n} \frac{3}{r^2+3r} &= \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+3}\right) \\ &= 1 - \frac{1}{4} + \frac{1}{2} - \frac{1}{5} + \frac{1}{3} - \frac{1}{6} + \frac{1}{4} - \frac{1}{7} + \cdots + \frac{1}{n} - \frac{1}{n+3} \\ &= 1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3} \\ &= \frac{11}{6} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3}.\end{aligned}$$

$$\sum_{r=1}^{n} \frac{1}{r^2+3r} = \frac{1}{3}\sum_{r=1}^{n} \frac{3}{r^2+3r} = \frac{1}{3}\left(\frac{11}{6} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3}\right).$$

9 b)

As $n \to \infty$, $\frac{1}{n+1} \to 0$, $\frac{1}{n+2} \to 0$ and $\frac{1}{n+3} \to 0$. Thus

$$\sum_{r=1}^{\infty} \frac{1}{r^2+3r} = \frac{11}{18}.$$

10)

$$\frac{1}{r^2-3r+2} = \frac{1}{(r-1)(r-2)} = \frac{A}{r-1} + \frac{B}{r-2},$$
$$1 = A(r-2) + B(r-1).$$

When $r = 1$, $1 = -A$; when $r = 2$, $1 = B$. Hence $A = -1$, $B = 1$ and $\frac{1}{r^2-3r+2} = \frac{1}{r-2} - \frac{1}{r-1}$.

$$\begin{aligned}\sum_{r=3}^{n} \frac{1}{r^2-3r+2} &= \sum_{r=3}^{n} \left(\frac{1}{r-2} - \frac{1}{r-1}\right) \\ &= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{n-3} - \frac{1}{n-2} + \frac{1}{n-2} - \frac{1}{n-1} \\ &= 1 - \frac{1}{n-1}.\end{aligned}$$

So $1 - \frac{1}{n-1} > 0.45 \Rightarrow \frac{1}{n-1} < 0.55$. Since the smallest value that n can be is $n = 3$, then $n - 1 > 0$, and so

$$1 < 0.55(n-1) \Rightarrow 1 < 0.55n - 0.55 \Rightarrow n > \frac{1.55}{0.55}.$$

Therefore $n = 3$.

11 a)

$$\frac{1}{(3r-2)(3r+1)} = \frac{A}{3r-2} + \frac{B}{3r+1},$$
$$1 = A(3r+1) + B(3r-2).$$

When $r = \frac{2}{3}$, $1 = 3A$; when $r = -\frac{1}{3}$, $1 = -3B$. Therefore $A = \frac{1}{3}$, $B = -\frac{1}{3}$ and

$$\frac{1}{(3r-2)(3r+1)} = \frac{1}{3(3r-2)} - \frac{1}{3(3r+1)}.$$

$$\begin{aligned}
\sum_{r=n}^{2n} \frac{1}{(3r-2)(3r+1)} &= \sum_{r=n}^{2n} \left(\frac{1}{3(3r-2)} - \frac{1}{3(3r+1)} \right) \\
&= \frac{1}{3(3n-2)} - \frac{1}{3(3n+1)} + \frac{1}{3(3n+1)} - \frac{1}{3(3n+4)} + \cdots \\
&\quad + \frac{1}{3(6n-2)} - \frac{1}{3(6n+1)} \\
&= \frac{1}{3(3n-2)} - \frac{1}{3(6n+1)} \\
&= \frac{(6n+1)-(3n-2)}{3(3n-2)(6n+1)} = \frac{n+1}{(3n-2)(6n+1)}.
\end{aligned}$$

11 b)

$$\begin{aligned}
\lim_{n\to\infty} \left(\sum_{r=n}^{2n} \frac{1}{(3r-2)(3r+1)} \right) &= \lim_{n\to\infty} \left(\frac{n+1}{(3n-2)(6n+1)} \right) \\
&= \lim_{n\to\infty} \left(\frac{n+1}{18n^2-9n-2} \right) \\
&= \lim_{n\to\infty} \left(\frac{1/n + 1/n^2}{18 - 9/n - 2/n^2} \right) = 0.
\end{aligned}$$

12 a)

$$\frac{15}{r^2+6r+8} = \frac{15}{(r+2)(r+4)} = \frac{A}{r+2} + \frac{B}{r+4},$$
$$15 = A(r+4) + B(r+2).$$

When $r = -4$, $15 = -2B$; when $r = -2$, $15 = 2A$. Hence $A = \frac{15}{2}$, $B = -\frac{15}{2}$, and

$$\frac{15}{r^2+6r+8} = \frac{15}{2(r+2)} - \frac{15}{2(r+4)}.$$

$$\begin{aligned}
\sum_{r=1}^{n} \frac{15}{r^2+6r+8} &= \frac{15}{2} \sum_{r=1}^{n} \left(\frac{1}{r+2} - \frac{1}{r+4} \right) \\
&= \frac{15}{2} \left(\frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \cdots + \frac{1}{n+2} - \frac{1}{n+4} \right)
\end{aligned}$$

$$
\begin{aligned}
&= \frac{15}{2}\left(\frac{1}{3}+\frac{1}{4}-\frac{1}{n+3}-\frac{1}{n+4}\right)\\
&= \frac{35}{8}-\frac{15}{2(n+3)}-\frac{15}{2(n+4)}\\
&= \frac{35n^2+125n}{8(n+3)(n+4)}.
\end{aligned}
$$

12 b)

As $n\to\infty$, $\frac{15}{2(n+3)}\to 0$ and $\frac{15}{2(n+4)}\to 0$. Therefore

$$\sum_{r=1}^{\infty}\frac{15}{r^2+6r+8}=\frac{35}{8}.$$

So that there is 4.375 million litres of water left in the long term.

12 c)

$$\sum_{r=1}^{30}1=30\Rightarrow\frac{35}{8}\times\sum_{r=1}^{30}\frac{1}{30}=\frac{35}{8}\Rightarrow\sum_{r=1}^{30}\frac{7}{48}=\frac{35}{8}.$$

$$\therefore\quad\sum_{r=1}^{n}\frac{15}{r^2+6r+8}-\sum_{r=1}^{30}\frac{7}{48}.$$

13 a)

$$
\begin{aligned}
\frac{300}{(r+1)(r+4)}&=\frac{A}{r+1}+\frac{B}{r+4},\\
300&=A(r+4)+B(r+1).
\end{aligned}
$$

When $r=-1$, $300=3A$; when $r=-4$, $300=-3B$. Thus $A=100$, $B=-100$ and

$$\frac{300}{(r+1)(r+4)}=\frac{100}{r+1}-\frac{100}{r+4}.$$

$$
\begin{aligned}
\sum_{r=1}^{n}\frac{300}{(r+1)(r+4)}&=100\sum_{r=1}^{n}\left(\frac{1}{r+1}-\frac{1}{r+4}\right)\\
&=100\left(\frac{1}{2}-\frac{1}{5}+\frac{1}{3}-\frac{1}{6}+\frac{1}{4}-\frac{1}{7}+\cdots+\frac{1}{n+1}-\frac{1}{n+4}\right)\\
&=100\left(\frac{1}{2}+\frac{1}{3}+\frac{1}{4}-\frac{1}{n+2}-\frac{1}{n+3}-\frac{1}{n+4}\right)\\
&=100\left(\frac{13}{12}-\frac{1}{n+2}-\frac{1}{n+3}-\frac{1}{n+4}\right)\\
&=\frac{25n(13n^2+81n+122)}{3(n+2)(n+3)(n+4)}.
\end{aligned}
$$

13 b)

As $n \to \infty$, $\frac{1}{n+2} \to 0$, $\frac{1}{n+3} \to 0$ and $\frac{1}{n+4} \to 0$. Therefore

$$\sum_{r=1}^{\infty} \frac{300}{(r+1)(r+4)} = 100 \times \frac{13}{12} = \frac{325}{3} = 108.3\ldots$$

So the sum in the long term will never exceed the £109 million as claimed.

13 c)

$$A_\infty = 100 \times \frac{13}{12} = \frac{325}{3} \text{ million.}$$

When $n = 30$:

$$100\left[\frac{13}{12} - \frac{1}{32} - \frac{1}{33} - \frac{1}{34}\right] = \frac{445375}{4488} = 99.2 \text{ million.}$$

$$\frac{445375/4488}{325/3} \times 100 = 91.6\% > 90\%.$$

The claim is verified that at least 90% of the total funding has been released.

14 a)

Since $\sin(A - B) = \sin A \cos B - \cos A \sin B$, then

$$\sin r \cos(r+1) - \cos r \sin(r+1) = \sin(r - (r+1)) = \sin(-1).$$

So

$$\begin{aligned} C_r &= \frac{\sin 1}{\cos r \cos(r+1)} = -\frac{\sin r \cos(r+1) - \cos r \sin(r+1)}{\cos r \cos(r+1)} \\ &= -\frac{\sin r \cos(r+1)}{\cos r \cos(r+1)} + \frac{\cos r \sin(r+1)}{\cos r \cos(r+1)} \\ &= -\tan r + \tan(r+1) = \tan(r+1) - \tan r, \quad \text{as req.} \end{aligned}$$

14 b)

$$C_r = \frac{\sin(-1)}{\cos r \sin(r+1)} = \tan(r+1) - \tan r.$$

$$\begin{aligned} \sum_{r=1}^{n} C_r &= \sum_{r=1}^{n} (\tan(r+1) - \tan r) \\ &= \tan 2 - \tan 1 + \tan 3 - \tan 2 + \tan 4 - \tan 3 + \cdots \\ &\quad + \tan n - \tan(n-1) + \tan(n+1) - \tan n \\ &= \tan(n+1) - \tan 1. \end{aligned}$$

15 a)

Since $\sinh(A - B) = \sinh A \cosh B - \cosh A \sinh B$, then

$$\sinh(r+1)\cosh r - \cosh(r+1)\sinh r = \sinh 1.$$

Therefore

$$\begin{aligned} D_r &= \frac{\sinh 1}{\sinh(r)\sinh(r+1)} \\ &= \frac{\sinh(r+1)\cosh r - \cosh(r+1)\sinh r}{\sinh(r)\sinh(r+1)} \\ &= \frac{\sinh(r+1)\cosh r}{\sinh(r)\sinh(r+1)} - \frac{\cosh(r+1)\sinh r}{\sinh(r)\sinh(r+1)} \\ &= \coth r - \coth(r+1). \end{aligned}$$

15 b)

$$D_r = \frac{\sinh 1}{\sinh(r)\sinh(r+1)} = \coth r - \coth(r+1).$$

$$\begin{aligned} \sum_{r=1}^{n} D_r &= \sum_{r=1}^{n} (\coth r - \coth(r+1)) \\ &= \coth 1 - \coth 2 + \coth 2 - \coth 3 + \coth 3 - \coth 4 + \cdots \\ &\quad + \coth(n-1) - \coth n + \coth n - \coth(n+1) \\ &= \coth 1 - \coth(n+1). \end{aligned}$$

16 a)

$$\tan((r+1)x - rx) = \tan x$$

and

$$\tan((r+1)x - rx) = \frac{\tan((r+1)x) - \tan(rx)}{1 + \tan((r+1)x)\tan(rx)}.$$

So

$$\begin{aligned} \tan x &= \frac{\tan((r+1)x) - \tan(rx)}{1 + \tan((r+1)x)\tan(rx)}, \\ 1 + \tan((r+1)x)\tan(rx) &= \frac{\tan((r+1)x) - \tan(rx)}{\tan x}, \\ \tan((r+1)x)\tan(rx) &= \frac{\tan((r+1)x)}{\tan x} - \frac{\tan(rx)}{\tan x} - 1, \quad \text{as req.} \end{aligned}$$

16 b)

Let $x = \frac{\pi}{20}$ so that

$$\begin{aligned} &\tan\frac{\pi}{20}\tan\frac{2\pi}{20} + \tan\frac{2\pi}{20}\tan\frac{3\pi}{20} + \cdots + \tan\frac{14\pi}{20}\tan\frac{15\pi}{20} \\ &= \sum_{r=1}^{14} \tan\left(\frac{\pi}{20}r\right)\tan\left((r+1)\frac{\pi}{20}\right) \\ &= \sum_{r=1}^{14} \left(\frac{\tan((r+1)\pi/20)}{\tan(\pi/20)} - \frac{\tan(r\pi/20)}{\tan(\pi/20)} - 1\right) \\ &= \frac{\tan(2\pi/20)}{\tan(\pi/20)} - \frac{\tan(\pi/20)}{\tan(\pi/20)} - 1 + \frac{\tan(3\pi/20)}{\tan(\pi/20)} - \frac{\tan(2\pi/20)}{\tan(\pi/20)} - 1 + \cdots \end{aligned}$$

$$
\begin{aligned}
&+\frac{\tan(15\pi/20)}{\tan(\pi/20)}-\frac{\tan(14\pi/20)}{\tan(\pi/20)}-1\\
&=-\frac{\tan(\pi/20)}{\tan(\pi/20)}+\frac{\tan(15\pi/20)}{\tan(\pi/20)}-14\\
&=\frac{\tan(15\pi/20)}{\tan(\pi/20)}-15.
\end{aligned}
$$

17 a)

$$
\begin{aligned}
u_{r+1}-u_r=\ln(r+2)-\ln(r+1)&=\ln\left(\frac{r+2}{r+1}\right)\\
&=\ln\left(\frac{r+1+1}{r+1}\right)=\ln\left(1+\frac{1}{r+1}\right).
\end{aligned}
$$

17 b)

The mean value theorem says that if f is continuous on $[a,b]$, then there exists $c\in(a,b)$ such that

$$\int_a^b f(x)\,dx=f(c)(b-a).$$

$$\int_{r+1}^{r+2}\frac{1}{x}\,dx=[\ln x]_{r+1}^{r+2}=\ln(r+2)-\ln(r+1).$$

By the mean value theorem, there exists a $c\in(r+1,r+2)$ such that

$$\int_{r+1}^{r+2}\frac{1}{x}\,dx=\frac{1}{c}((r+2)-(r+1))=\frac{1}{c}.$$

Hence $u_{r+1}-u_r=\frac{1}{c}$, where $c\in(r+1,r+2)$. Since $r+1<c<r+2$, then

$$\frac{1}{r+2}<\frac{1}{c}<\frac{1}{r+1}.$$

Therefore $\frac{1}{r+2}<u_{r+1}-u_r<\frac{1}{r+1}$, as req.

17 c)

$$
\begin{aligned}
\sum_{r=1}^{n}(u_{r+1}-u_r)&=u_2-u_1+u_3-u_2+u_4-u_3+\cdots+u_n-u_{n-1}+u_{n+1}-u_n\\
&=u_{n+1}-u_1=\ln(n+1+1)-\ln(1+1)\\
&=\ln(n+2)-\ln 2.
\end{aligned}
$$

18 a)

$$
\begin{aligned}
\frac{1}{x(x+1)}&=\frac{A}{x}+\frac{B}{x+1},\\
1&=A(x+1)+Bx.
\end{aligned}
$$

When $x=0$, $1=A$; when $x=-1$, $1=-B$. Hence $A=1$, $B=-1$, and $\frac{1}{x(x+1)}=\frac{1}{x}-\frac{1}{x+1}$.

$$y=\int\left(\frac{1}{x}-\frac{1}{x+1}\right)dx=\ln x-\ln(x+1)+c=\ln\left(\frac{x}{x+1}\right)+c.$$

18 b)

$$u_{r+1} - u_r = \ln\left(\frac{r+1}{r+2}\right) - \ln\left(\frac{r}{r+1}\right).$$

$$\begin{aligned}\text{Alternatively: } u_{r+1} - u_r &= (\ln(r+1) - \ln(r+2) + c) - (\ln r - \ln(r+1) + c)\\ &= -\ln r + 2\ln(r+1) - \ln(r+2).\end{aligned}$$

18 c)

$$\begin{aligned}\sum_{r=1}^{n}(u_{r+1} - u_r) &= \sum_{r=1}^{n}\left(\ln\left(\frac{r+1}{r+2}\right) - \ln\left(\frac{r}{r+1}\right)\right)\\ &= \ln\frac{2}{3} - \ln\frac{1}{2} + \ln\frac{3}{4} - \ln\frac{2}{3} + \ln\frac{4}{5} - \ln\frac{3}{4} + \cdots + \ln\frac{n+1}{n+2} - \ln\frac{n}{n+1}\\ &= \ln\left(\frac{n+1}{n+2}\right) - \ln\left(\frac{1}{2}\right) = \ln\left(\frac{2(n+1)}{n+2}\right).\end{aligned}$$

Alternatively,

$$\begin{aligned}\sum_{r=1}^{n}(u_{r+1} - u_r) &= \sum_{r=1}^{n}(-\ln r + 2\ln(r+1) - \ln(r+2))\\ &= -\ln 1 + 2\ln 2 - \ln 2 - \ln(n+1) + 2\ln(n+1) - \ln(n+2)\\ &= \ln 2 + \ln(n+1) - \ln(n+2) = \ln\left(\frac{2(n+1)}{n+2}\right).\end{aligned}$$

18 d)

$$\sum_{r=1}^{\infty}(u_{r+1} - u_r) = \lim_{n\to\infty}\ln\left(\frac{2(n+1)}{n+2}\right) = \ln 2.$$